Project Topic Proposal

CCTV Control System for University Campus

AI-Assisted University CCTV Monitoring and Violence Prevention System

Team members

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Proposal summary

- **Problem:** Security staff cannot watch many camera feeds at once, so fights on campus are noticed late or found only afterwards in recorded footage.
- **Solution:** The system reads live camera streams, detects fights automatically with an AI model, saves a short evidence clip, and sends a real-time alert to a web dashboard.
- **Human decision:** A security operator confirms or dismisses each alert. Unacknowledged alerts escalate to a supervisor after 60 seconds.

User roles

- **Security operator:** verifies alerts and reviews evidence clips.
- **Security supervisor:** handles escalations, reviews incident history and reports.
- **System administrator:** manages cameras, users, detection thresholds and data retention.

Competing quality attributes

- **Speed vs. accuracy:** bigger models detect better but are slower; lower thresholds catch more fights but raise false alarms.
- **Security vs. ease of use:** access control, encryption and audit logs protect privacy but add steps for operators.

External integrations

- **Claude AI API:** external system connected to the CCTV Control System (two-way, as in the C4 context diagram); uploaded videos are stored persistently, not in temporary application memory.
- **University CCTV cameras:** external system that provides live video streams (RTSP/ONVIF; recorded videos in the prototype).

Separated, independently deployable parts

- **Frontend:** React web dashboard for operators, supervisors and administrators.
- **Backend services:** incident/alert service, ingestion service and AI detection service, linked by a message queue, each in its own container.

Scope

- Ingest live video streams from registered cameras.
- Detect fights automatically in camera footage.
- Face recognition.
- Save a short evidence clip with camera ID, timestamp and confidence for every detection.
- Real-time alert on the dashboard when a possible fight is detected.
- Confirm or dismiss an alert, with an optional comment.
- Escalate to the supervisor if an alert is not acknowledged within 60 seconds.
- Delete footage and evidence automatically after a configurable retention period.
- Access control, encryption and audit logs for evidence.

Milestone 1
Proposal & Architectural Requirements

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Contribution table

Section	Responsible member
1. System overview	Hussien Hesham /Mariam Wael
2. Stakeholders and concerns	Hussien Hesham
3. System context (C4 Level 1 diagram)	Mario Raffat / Abdelrahman tamer
4. Functional requirements	Mario Raafat
5. Quality attribute scenarios	Abdelrahman tamer
6. Constraints	Zeina Mohamed
7. Architecturally significant requirements	Mariam Wael

I. System Overview

The problem. University campuses are open, crowded environments where physical fights can break out without warning. Security staff must watch many camera feeds at once, and human attention degrades quickly over long shifts, so incidents are often noticed late or only discovered afterwards by manually reviewing recorded footage. Delayed detection means delayed intervention.

Who uses it. *Security operators* (verify alerts and review clips), *security supervisors* (monitor incidents, handle escalations, review reports), *system administrators* (manage cameras, users, detection thresholds and retention).

Why the System Is Architecturally Interesting

The CCTV control system for university campuses is more than a dashboard that shows videos. It must watch live cameras around the clock, spot a fight within seconds, and do it while protecting people's privacy. That mix makes the design choices important.

It never stops working, and the work is heavy. Live videos must be decoded and analyzed by an AI model almost in real time. We can't let one slow step hold up everything else, so we separate the stages with queues, each stage working on its own and passing results on. We also need to decide how to share the GPU and CPU workers. And we need a plan for when video arrives faster than we can analyses it, so the system slows down safely instead of crashing. This is called back pressure.

Good qualities compete with each other. Improving one often costs us another:

Speed vs. accuracy: Bigger models are more accurate but slower. Lower detection thresholds catch more fights, but they also raise more false alarms.

Security vs. ease of use: Access control, encryption and audit logs protect students, but they add extra steps and a little delay.

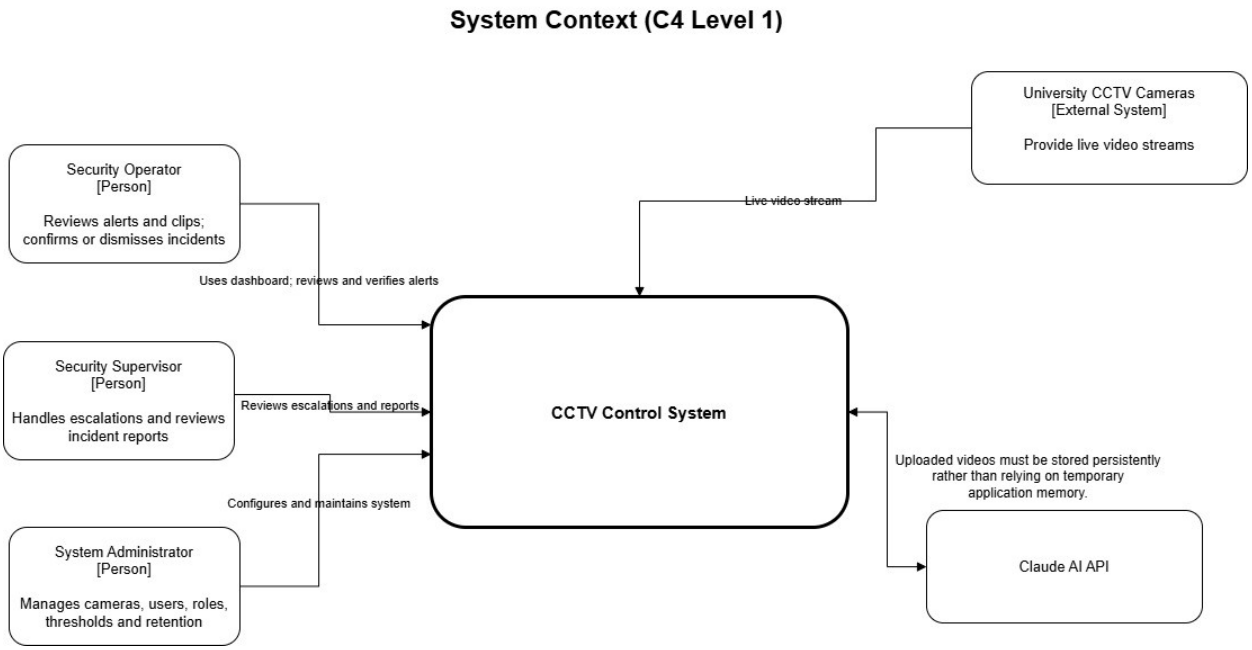
Easy to change vs. fast: Keeping the model easy to swap makes the system flexible, but a tightly tuned pipeline would run faster.

2. Stakeholders and Concerns

Stakeholders were identified from the people who use, operate, fund, regulate, build or are affected by the system.

#	Stakeholder	Role	Key concerns
S1	Security operators	Primary end users; watch the dashboard and verify alerts	Alerts arrive fast and are understandable; few false alarms . simple confirm/dismiss workflow; clip shows enough context to decide in seconds
S2	security manager	Oversee operators, handle escalations, request reports	Reliable escalation when alerts are missed; incident history and reports; camera coverage; confidence that the system is working (availability)
S3	business owner	Sponsor; accountable for campus safety	Improved safety and response time; zero or low cost (open-source stack, existing cameras); legal and reputational risk; demonstrable evaluation results
S4	administrators	Deploy, configure and maintain the system	Easy deployment and monitoring; recovery from failures; adding cameras; managing users and roles; limited hardware
S5	Students, staff and visitors	Indirect users; their images are processed	Privacy and proportionality; footage not misused or leaked; fair treatment
S6	developers	Build, test and evolve the system	Modularity; ability to swap or retrain models; testability with recorded videos; clear interfaces; maintainability within a 5-person team

3. System Context (C4 Level 1)



4.Functional Requirements

User stories are prioritised with MoSCoW (Must / Should / Could / Won't for this release). The last column marks whether the Milestone 3 architectural prototype will implement the story. The prototype deliberately covers one complete end-to-end flow (ingest → detect → store evidence → alert → verify) with security, rather than many shallow features.

ID	User story	MoSCoW	M3 prototype
FR-01	As the **system**, I ingest live video streams from registered cameras so that footage can be analysed continuously.	Must	Yes
FR-02	As a **security operator**, I want fights in camera footage detected automatically so that I do not have to watch every feed.	Must	Yes
FR-03	As a **security operator**, I want a short evidence clip (with camera ID, timestamp, confidence) saved for every detection so that I can verify what happened.	Must	Yes
FR-04	As a **security operator**, I want a real-time alert on the dashboard when a possible fight is detected so that I can react immediately.	Must	Yes
FR-05	As a **security operator**, I want to confirm or dismiss an alert (with optional comment) so that a human decides and false alarms are recorded.	Must	Yes
FR-06	As a **security supervisor**, I want another alert when an alert is not acknowledged within 60 s so that no incident is missed.	must	Yes
FR-07	As a **system administrator**, I want footage and evidence deleted automatically after a configurable retention period so that data is not kept longer than needed.	Should	Yes
FR-8	As a **security operator**, I want each alert ranked low / medium / high severity so that urgent incidents are handled first.	Should	No
FR-9	As a **system administrator**, I want to register cameras, enable/disable them and see their health status so that coverage is maintained.	Should	No
FR-10	As a **security supervisor**, I want to search and filter incident history and export a report so that I can review patterns.	could	No
FR-11	As a **system administrator**, I want a performance dashboard (precision, false-alarm rate, latency) so that I can tune the system.	should	No

5.Quality Attribute Scenarios

Seven scenarios in the six-part form (source, stimulus, artifact, environment, response, response measure) [1] cover seven quality attributes. Numeric targets are initial values chosen to be testable.

QAS-1 — Performance (detection latency)	
Source	A campus camera (live stream)
Stimulus	Sporadic (Unpredictable, No pattern), a physical fight becomes visible in the camera view
Artifact	Ingestion service, detection service and incident/alert service (the detection pipeline)
Environment	Normal operation; 10 cameras analysed at 10 fps on the single available server
Response	The fight is detected, an evidence clip is stored, and the alert appears on all logged-in operator dashboards
Response measure	For 95% of fights, the alert is shown within 8 s of fight onset (5 s analysis window + ≤ 3 s processing and delivery), measured over 100 replayed test incidents

QAS-2 — Availability	
Source	Internal fault (software crash or out-of-memory error)
Stimulus	A detection-service instance crashes while analysing streams
Artifact	Detection service and message queue
Environment	Normal operation; 10 cameras streaming 24/7
Response	The failure is detected automatically, the instance is restarted, and queued frames/clips are processed after recovery; operators are notified of the degraded state
Response measure	Service restored in ≤ 30 s; no more than 60 s of video left unanalysed after recovery; alerting pipeline availability ≥ 99% over a 7-day test run. Availablility = (MTTF / (MTTF + MTTR)).

QAS-3 — Security (confidentiality and auditability)	
Source	An authenticated user without the required role, or an unauthenticated client
Stimulus	Attempts to view or export an evidence clip or call an admin API
Artifact	Incident API and evidence store
Environment	Normal operation, over the campus network
Response	The request is rejected (HTTP 401/403), the attempt is logged, and the supervisor is notified after 5 failed attempts within 10 minutes
Response measure	100% of 50 scripted unauthorized requests are denied; 100% of permitted views/exports appear in the audit log with user ID and timestamp; evidence encrypted at rest (AES-256) .

QAS-4 — Modifiability	
Source	ML engineer
Stimulus	Wants to replace the fight-detection model with a newer model version
Artifact	Detection service (model adapter interface)
Environment	Design / Deployment time.
Response	The new model is plugged in behind the same interface and deployed without changing ingestion, API or dashboard code
Response measure	≤ 1 person-day of effort; changes confined to ≤ 2 files in the detection service; 0 changes to other services; regression test suite passes

QAS-5 — Scalability	
Source	System administrator
Stimulus	Campus adds cameras, growing from 10 to 50 (5× load)
Artifact	Ingestion service and detection service (worker pool)
Environment	Normal operation at peak hours
Response	Capacity is increased by adding detection workers and registering cameras, without redesign or downtime of existing streams
Response measure	At ≤ 5× the compute resources, all 50 cameras are analysed at 10 fps with QAS-1 (p95 ≤ 8 s) still met; scaling performed by configuration in ≤ 1 hour

QAS-6 — Usability	
Source	Security operator
Stimulus	An alert arrives while up to 5 other alerts are open
Artifact	Web dashboard
Environment	Normal operation
Response	The dashboard highlights the new alert, auto-plays the evidence clip with camera location and confidence, and offers one-click confirm / dismiss
Response measure	Median time to verify an alert ≤ 20 s using ≤ 3 clicks; System Usability Scale score ≥ 75 with at least 5 test users

QAS-7 — Testability	
Source	Developer / tester
Stimulus	Runs an end-to-end test using recorded, labelled videos instead of live cameras
Artifact	Ingestion service and detection pipeline
Environment	Test environment, no physical cameras
Response	Video files are replayed through the same interface as a camera stream; alerts and metrics (precision, recall, latency) are computed automatically
Response measure	A 200-clip test set runs in ≤ 30 min; 3 repeated runs give identical labels (100% repeatability); ≥ 70% unit-test coverage of core services

6.Constraints

Constraints are limits that we must accept. We did not choose them. They come from the technology we can use, the money we have, and the course rules. Reference numbers [1]–[10] are from the reference list of the main document.

Technical constraints

ID	Constraint	impact
T1	We use Python for the AI part (OpenCV, PyTorch or TensorFlow).	The team already knows Python and it has the best AI libraries.
T2	Real cameras send video with RTSP/ONVIF. In the prototype we use recorded videos instead of real cameras.	We cannot use the real campus cameras during the project.
T3	We have only one normal computer with one GPU (or CPU only). We use the cloud only to train the model, and only with public datasets.	The computer has limited power, so we plan for about 10 cameras (QAS-1, QAS-5).
T4	The public datasets are for research only and may need a signed agreement.	We cannot use them commercially or upload the videos to GitHub .
T5	Each part of the system runs as a container. We use free tools: Fast API/Flask, React and MySQL.	Each part can be deployed on its own, and there is no license cost.
T6	Users log in with the university system . In the prototype we use a free test login .	We need it for user roles and access control (QAS-3).
T7	The dashboard is a React website for desktop browsers. There is no mobile app.	It keeps the work small enough for 5 students (FR-04, FR-05, QAS-6).

Business constraints

ID	Constraint	Rationale / impact
B1	The system only sends alerts. A person always decides what to do.	Wrong alerts cause less harm, and we need a confirm/dismiss step (FR-05).
B2	face recognition. The system identifies people.	It is a legal and ethical rule .
B3	We must follow the Personal Data Protection Law: access control, activity logs, deleting old data, and a notice. CCTV in public places needs a licence.	It affects FR-07 and QAS-3. We use public datasets only .
B4	We test accuracy only on public datasets.	We cannot say it works on the real campus (follows from T4 and O4).

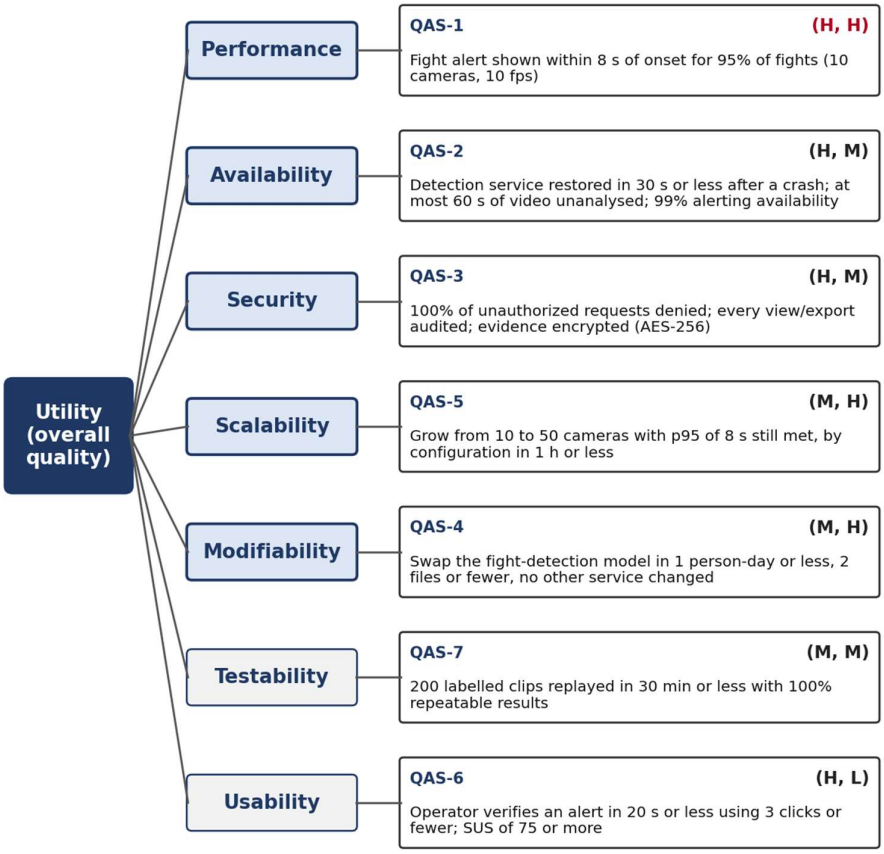
Organizational constraints

ID	Constraint	Rationale / impact
O1	The team has exactly 5 students and fixed deadlines for milestones M1 to M4.	The prototype covers only one full flow from start to end.
O2	We can use real campus videos only with permission from the university.	We cannot count on having them, so we use public datasets.
O3	Every design decision is written as an ADR and linked to an important requirement.	It helps us with the ATAM evaluation in M4.

7. Architecturally Significant Requirements (ASRs)

The requirements from Sections 4–6 were expressed as seven quality attribute scenarios (QAS-1 to QAS-7). Each was rated for business value and architectural impact. The ratings are shown in the utility tree and ranked in the table below; the top five are justified as the ASRs.

7.1 Utility Tree



Leaf label = (Business value, Architectural impact). Shaded attributes = top 5 ASRs.

7.2 Justification of the Top 5 ASRs

#	ASR	Why it shapes the architecture
1	**Performance (QAS-1)**	Highest in both value and impact. Early fight detection is the core purpose of the system, and a 3 s processing time.
2	**Availability (QAS-2)**	A monitoring system that is down misses' incidents. Recovery within 30 s without losing video cannot be added late; it requires buffering and supervision from the start.
3	**Security (QAS-3)**	Protecting student privacy is a stated objective and a Must feature (audit and privacy controls). Access control and encryption affect every service that touches footage.
4	**Scalability (QAS-5)**	Growth from 10 to 50 cameras (5x load) must not require redesign.
5	**Modifiability (QAS-4)**	Violence-detection models change quickly, so the model is the most likely part to be replaced.

8. References

[1] https://www.researchgate.net/publication/390583055_Enhancing_Public_Safety_with_AI_ML-Based_CCTV_Surveillance

[2] https://www.researchgate.net/publication/391477868_Detection_of_Criminal_Activities_and_Anomalies_through_CCTV's

[3] https://ieeexplore.ieee.org/document/9012234/metrics#metrics